

Let $C(x)$ be the statement “ x has a cat”, let $D(x)$ be the statement “ x has a dog”, and let $F(x)$ be the statement “ x has a ferret”. Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

A student in your class has a cat, a dog, and a ferret.

All students in your class have a cat, a dog, or a ferret.

Some student in your class has a cat and a ferret, but not a dog.

No student in your class has a cat, a dog, and a ferret.

For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

Let $P(x)$ be the statement “ x can speak Russian” and let $Q(x)$ be the statement “ x knows the computer language C++”. Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

There is a student at your school who can speak Russian and who knows C++.

There is a student at your school who can speak Russian but who doesn’t know C++.

Every student at your school either can speak Russian or knows C++.

No student at your school can speak Russian or knows C++.

Translate these statements into English, where the domain for each variable consists of all *Real* numbers.

$$\forall x \exists y (x < y)$$

$$\forall x \forall y (((x \geq 0) \wedge (y \geq 0)) \rightarrow (xy \geq 0))$$

$$\forall x \forall y \exists z (xy = z)$$

$$\exists x \forall y (xy = y)$$

$$\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$$

$$\forall x \forall y \exists z (x = y + z)$$

Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

$$\exists x \exists y Q(x, y)$$

$$\exists x \forall y Q(x, y)$$

$$\forall x \exists y Q(x, y)$$

$$\exists y \forall x Q(x, y)$$

$$\forall y \exists x Q(x, y)$$

$$\forall x \forall y Q(x, y)$$

Let $T(x, y)$ mean that student x likes cuisine y , where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.

$\neg T(\text{Abdallah Hussein}, \text{Japanese})$

$\exists x T(x, \text{Korean}) \wedge \forall x T(x, \text{Mexican})$

$\exists y (T(\text{Monique Arsenault}, y) \vee T(\text{Jay Johnson}, y))$

$\forall x \forall z \exists y ((x \neq z) \rightarrow \neg (T(x, y) \wedge T(z, y)))$

$\exists x \exists z \forall y (T(x, y) \leftrightarrow T(z, y))$

$\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$

Let $S(x)$ be the predicate “ x is a student,” $F(x)$ the predicate “ x is a faculty member,” and $A(x, y)$ the predicate “ x has asked y a question,” where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.

Lois has asked Professor Michaels a question.

Every student has asked Professor Gross a question.

Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.

Some student has not asked any faculty member a question.

There is a faculty member who has never been asked a question by a student.

Some student has asked every faculty member a question.

There is a faculty member who has asked every other faculty member a question.

Some student has never been asked a question by a faculty member.

Express each of these mathematical statements using predicates, quantifiers, logical connectives, and mathematical operators.

The product of two negative real numbers is positive.

The difference of a real number and itself is zero.

Every positive real number has exactly two square roots.

A negative real number does not have a square root that is a real number.

Express the negations of each of these statements so that all negation symbols immediately precede predicates.

$\forall x \exists y \forall z T(x, y, z)$

$\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$

$\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$

$\forall x \exists y (P(x, y) \rightarrow Q(x, y))$

Find a common domain for the variables x , y , and z for which the statement $\forall x \forall y ((x \neq y) \rightarrow \forall z ((z = x) \vee (z = y)))$ is true and another domain for which it is false.